Full-Stack Agricultural Marketplace

AgriLink

Connecting Farms Directly to Markets

A digital platform that connects verified farmers with buyers, improves price transparency, and manages the complete marketplace workflow.

Project Presentation

1. Problem Statement

Challenges for Farmers

- Dependence on intermediaries
- Unstable and unclear pricing
- Difficulty reaching direct buyers
- Limited digital inventory tools

Challenges for Buyers

- Limited information about product origin
- Inconsistent quality and availability
- Hidden or unclear charges
- Difficulty finding trusted farmers

AgriLink addresses these problems with one transparent, role-based online marketplace.

2. Proposed Solution

- Farmers create and manage produce listings.
- Buyers browse products and place orders directly.
- Farmer verification increases trust on the platform.
- Inventory, order status, pricing, and analytics are managed digitally.
- Administrators moderate users and marketplace content.

Project Objectives

Connect farmers directly with customers.

Improve farmer earnings and market access.

Provide transparent pricing and secure ordering.

Support data-driven marketplace decisions.

3. User Roles

Farmer

- Manage produce listings
- Upload verification documents
- Control inventory and prices
- Process orders
- View revenue analytics

Buyer

- Search and filter products
- View verified farmers
- Manage cart and checkout
- Track orders
- Submit reviews

Administrator

- Review farmer verification
- Manage users
- Moderate listings
- View GMV and commissions
- Monitor audit logs

4. Main Features

Marketplace

Search, category, price, location, quality-grade, farming-method, and verified-farmer filters.

Produce Management

Product name, category, price, unit, quantity, minimum order quantity, quality grade, farming method, location, and images.

Order Fulfillment

```
PLACED -> CONFIRMED -> PROCESSING -> READY_FOR_DELIVERY
       -> OUT_FOR_DELIVERY -> DELIVERED
```

Analytics

Farmers can view sales performance, while administrators can view marketplace GMV, platform commissions, active listings, and category statistics.

5. Transparent Pricing

The buyer sees the complete price before placing an order.

> Total Payable = Produce Subtotal + Platform Fee + Delivery Fee

Example

Component	Amount
Produce subtotal	Rs. 1,000
Platform fee: 3%	Rs. 30
Delivery fee	Rs. 70
Total payable	**Rs. 1,100**

This pricing model reduces confusion and builds trust between buyers and the platform.

6. Technology Stack

Frontend

- React 18
- Vite
- React Router
- Tailwind CSS
- Axios
- Recharts and Framer Motion

Backend

- Node.js
- Express.js
- Mongoose
- JWT authentication
- bcryptjs
- Helmet and rate limiting

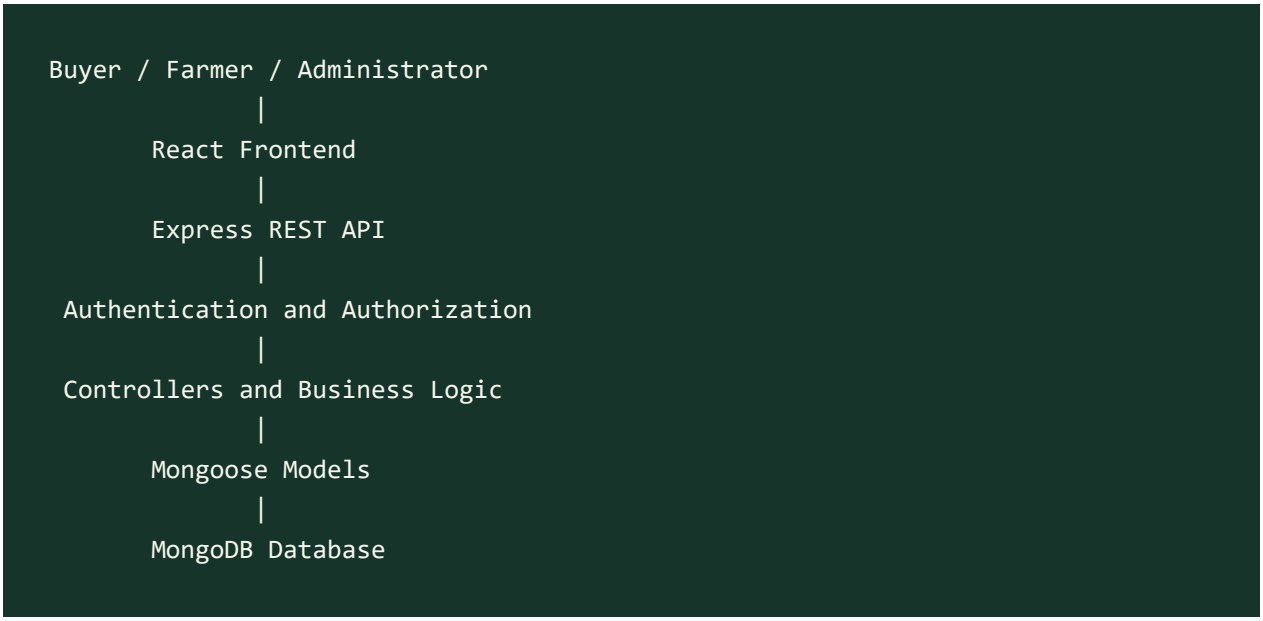
Database

- MongoDB
- User and role data
- Products and inventory
- Orders and reviews
- Notifications and audit logs

Development

- REST API architecture
- Environment configuration
- Local development servers
- Seeded demo accounts

7. System Architecture



The frontend sends requests to the backend. The backend validates permissions, applies marketplace rules, and reads or writes data in MongoDB.

Running Services

Frontend: `http://localhost:5173`
Backend: `http://localhost:5000`
Database: `mongodb://127.0.0.1:27017/agrilink`

8. Backend API

Method	Endpoint	Purpose
POST	/api/auth/register	Register a user
POST	/api/auth/login	Authenticate a user
GET	/api/products	Browse and filter products
POST	/api/products	Create a product listing
GET	/api/cart	View the buyer cart
POST	/api/orders	Place an order
PATCH	/api/orders/:id/status	Update order status
GET	/api/farmer/analytics	View farmer analytics
GET	/api/admin/analytics	View platform analytics

9. Database Design

Collections

- Users
- Products
- Orders
- Cart
- Reviews
- Notifications
- Audit logs

Relationships

- A farmer owns product listings.
- A buyer creates orders.
- Orders contain product references.
- Buyers submit reviews.
- Important actions create audit records.

Live Connection

The application connects through the server environment variable:

```
MONGODB_URI=mongodb://127.0.0.1:27017/agrilink
```

The products API has been verified and returns MongoDB-backed product data.

10. Security

- JWT-based authentication
- Password hashing with bcryptjs
- Role-based authorization
- Protected farmer, buyer, and admin routes
- Helmet security headers
- API rate limiting
- Input and request validation
- Stock and minimum-order validation
- Audit logging for important actions

Why It Matters

These controls protect user accounts, prevent unauthorized actions, and help maintain reliable marketplace operations.

11. Live Demonstration

Buyer Demo

Log in, search products, apply filters, add an item to the cart, review the price breakdown, place an order, and track its status.

Farmer Demo

Log in, create or edit a listing, update stock, view incoming orders, change an order status, and review analytics.

Administrator Demo

Log in, review marketplace metrics, approve a farmer, manage users, moderate listings, and inspect audit activity.

Demo Accounts

Role	Email	Password
Admin	admin@agrilink.demo	Password123!
Farmer	farmer@agrilink.demo	Password123!
Buyer	buyer@agrilink.demo	Password123!

12. Advantages and Future Scope

Advantages

- Direct farmer-to-buyer commerce
- Transparent fees
- Verified farmer profiles
- Digital inventory and orders
- Role-based access
- Analytics and audit trails

Future Enhancements

- Online payment integration
- Live delivery tracking
- SMS and email notifications
- Mobile application
- AI-based price prediction
- Weather and crop advisory
- Cloud deployment

Conclusion

AgriLink is a full-stack agricultural marketplace that creates a direct digital bridge between farms and consumers.

It combines transparent pricing, verified users, inventory management, secure authentication, order tracking, and analytics in one platform.

Thank you